NOTES

TUT - 8.

2) → Prim's

```
Prim (x, y, z);
  For each vertex u ∈ x.V;
    key [u] = ∞
    parent [u] = NIL
    key [z] = 0
    Q = Min-Priority Queue (x.V)
    while Q ≠ 0
    u = extract-Min(Q)
  for each v ∈ Adj [u]
    if v ∈ Q & y(u, v) < key [v]
      parent [v] = u
      key [u] = w(u, v)
```

NOTES

Kruskal's

```
Kruskal [G, w)
  A = ∅
for each vertex v ∈ G.V
  Make-Set (v)
Sort G.E by non decresing weight
for each edge (u, v) ∈ G.E in sorted order
  if Find-Set (u) ≠ Find-Set (v)
      A = A ∪ {(u,v)}
      Union (u, v)
      if |A| = |V| - 1;
        break
Return A.
```

NOTES

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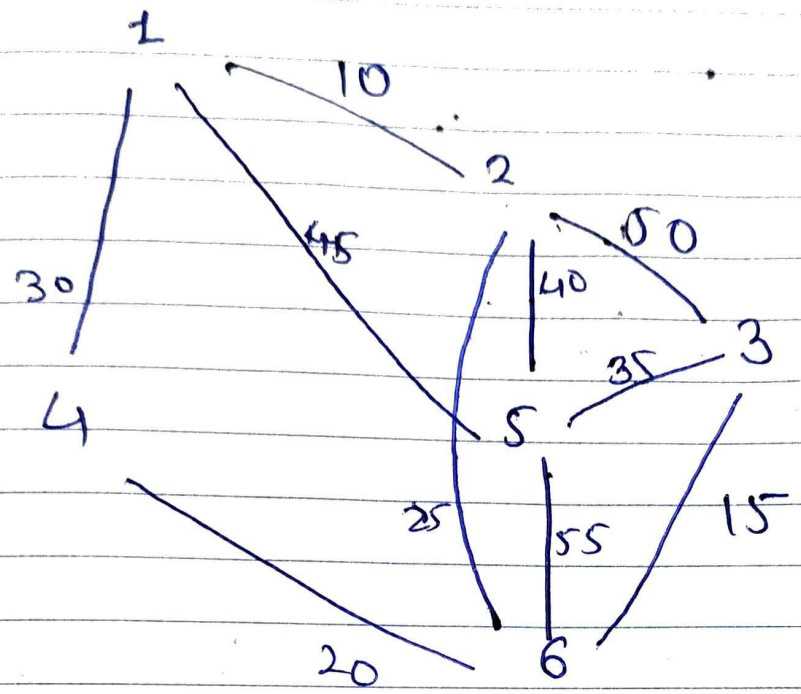
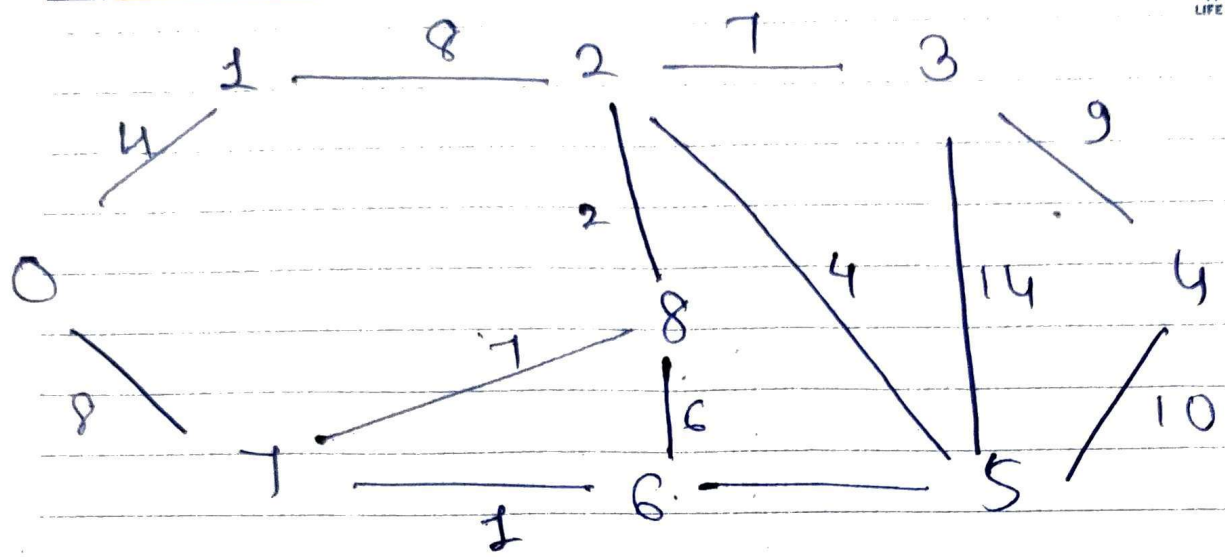
Q2.



Prim's	Edge	weight	Krukal's	Weight
1	1 → 2	10	1 → 2	10
2	1 → 4	30	3 → 6	15
3	4 → 6	20	4 → 6	20
4	6 → 3	15	4 → 5	25
5	4 → 5	25	1 → 4	30
		100		100

Sama

NOTES



Prim's	Weight	Krukal's	Weight
0-1	4	6-7	1
0→7	8	2-8	2
7→6	1	5-6	2
6→5	2	0-1	4
5→2	4	2→5	4
2→8	2	2→3	7
2→3	7	0-7	8
3→4	9	3→4	9
	11		11
	37		37

Same

Har Budget ke liye Sahi Policy

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3) Sort all by increasing finish time f[i]

For remaining

if s[i] >= last_selected_finish_time

Activity Selection (s, f, n)

sort activities by f[i] in ascending

A = { 1 }

k = 1

for i = 2 to n

if s[i] >= f[k];

~~if~~ add i to A

k = i

return A.